

arrays, which is nothing but support for Boolean indexing. The example given below explains the same.

```
>>>c=n.array([1,4,7,8,2])
>>>d=c<5
>>>d
array([ True,  True, False, False,  True], dtype=bool)
>>>c[d]
array([1, 4, 2])
```

Here, 'd' is a Boolean array whose element is set to 'True' if the corresponding element in 'c' has a value less than 5. Accessing array 'c' using 'd', i.e., `c[d]`, will fetch an element in 'c' only if the element in the corresponding index position in 'd' is 'True'. I will give one more example. An array 'a' is defined as follows:

```
>>>a=n.array([1,2,3,4,5,6,7])
```

We can see that the statement given below will retrieve all elements in array 'a' which are even numbers.

```
>>>a[a%2==0]
array([2, 4, 6])
```

Integer overflow in Python

Python 2 supports two types of integers: *int* and *long*. *Int* is C type, which allows a range of values to be taken, while *long* is arbitrary precision whose maximum value is limited by the available memory. If *int* is not enough, it will be automatically

promoted to *long*. When it comes to Python 3, there is support for arbitrary precision integers. So there is no question of overflow in integer operations in pure Python. But we cannot restrict our use to pure Python, since scientific computation needs packages in the PyData stack (e.g., NumPy, Pandas, SciPy, etc). The PyData stack uses C type integers which have fixed precision. It uses 64 bits for representation. So the maximum value an integer can take is $2^{63}-1$. The overflow condition is shown below:

```
>>>a=n.array([2**63-1,4],dtype=int)
>>>a
array([9223372036854775807,          4])
>>>a+1
array([-9223372036854775808,          5])
```

To conclude, NumPy not only makes computation easier, but also makes the program run faster. It provides multi-dimensional arrays and tools to play with arrays. **END** 

References

- [1] wikipedia.org
- [2] scipy.org
- [3] www.quora.com/What-is-the-relationship-among-NumPy-SciPy-Pandas-and-Scikit-learn-and-when-should-I-use-each-one-of-them

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It is one of the most popular Python based CMS frameworks, so people who are familiar with Python can try it out. There are many websites built using this CMS.

The official website for it is <https://www.django-cms.org/en/>.

10. Liferay

Liferay is a free and open source enterprise software product that is mainly used for corporate intranets and extranets. It has inbuilt CMS support and is written in Java. It has an open source and enterprise version. Liferay Portal CE is the open source version of Liferay's enterprise Web platform.

The official website for it is <https://www.liferay.com/>. **END** 

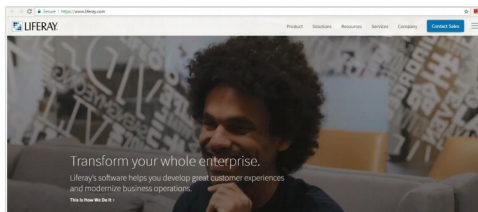


Figure 10: Liferay

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